Probabilistic Reasoning

Outline

- Probability theory - basics
- Probabilistic Inference
  - Inference by enumeration
- Conditional Independence
- Bayesian Networks
- Inference in Bayesian Networks
  - Exact Inference
  - Approximate Inference

Uncertainty

- The problem might not be in the fact that T/F values can change over time but rather that we are not certain of the T/F value
- Agents almost never have access to the whole truth about their environment
- Agents must act in the presence of uncertainty
  - Incompleteness and/or incorrectness of rules used by the agent
  - Limited and ambiguous sensors
  - Imperfection/noise in agent's actions
  - Dynamic nature of the environment

Pitfalls of pure logic

- Laziness
  - Too much work to list all conditions needed to ensure an exceptionless rule
- Theoretical ignorance
  - Science has no complete theory for the domain
- Practical ignorance
  - Even if we know all the rules, we may be uncertain about them
  - We only have a degree of belief in them

Probability Theory vs. Logic

- Probability - tool for handling degrees of belief
  - Summarizes uncertainty due to laziness and ignorance

	Ontological commitments	Epistemological commitments
Logic	world is composed of facts that do or do not hold	each sentence is true or false or unknown
Probability Theory	world is composed of facts that do or do not hold	numerical degree of belief between 0(sentence for certainly false) and 1(sentences are certainly true)

Rational Agent Approach

- Choose action A that maximizes expected utility
  - Maximizes Prob(A)*Utility(A)
- Prob(A) - probability that A will succeed
- Utility(A) - utility value to agent of A's outcomes

$$
\begin{aligned}
P(A_{25} \text{ gets me there on time} | \ldots) &= 0.04 \\
P(A_{90} \text{ gets me there on time} | \ldots) &= 0.70 \\
P(A_{120} \text{ gets me there on time} | \ldots) &= 0.95 \\
P(A_{1440} \text{ gets me there on time} | \ldots) &= 0.9999
\end{aligned}
$$

Which action to choose?

Depends on my preferences for missing flight vs. airport cuisine, etc.

Utility theory is used to represent and infer preferences

Decision theory = utility theory + probability theory

Propositions

- Think of a proposition as the event (set of sample points) where the proposition is true
- Given Boolean r.v. A and B:
  - Event a = set of sample points where $A(\omega) = true$
  - Event $\neg a$ = set of sample points where $A(\omega) = false$
  - Event $a \wedge b$ = sample points were $A(\omega) = true$ and $B(\omega) = true$
- Often in AI applications, the sample points are defined by the values of a set of random variables i.e., the sample space is the Cartesian product of the ranges of the variables
- With Boolean variables, sample point is propositional logic model
  - $A = true, B = false, a \vee \neg b$
- Proposition is disjunctions of atomic events in which it is true
  - $(a \vee b) = (a \wedge b) \vee (a \wedge \neg b) \vee (\neg a \wedge b)$
  - $\Rightarrow P(a \vee b) = P(a \wedge b) + P(a \wedge \neg b) + P(\neg a \wedge b)$

Syntax for Propositions

- Propositional or Boolean random variables
  - Example: Cavity (do I have a cavity?)
  - Cavity = true is a proposition, also written as $cavity$
- Discrete random variables (finite or infinite)
  - Example: Weather is one of {sunny, rain, cloudy, snow}
  - $Weather = rain$ is a proposition
  - Values must be exhaustive and mutually exclusive
- Continuous random variables
  - Example: Temperature = 37
- Arbitrary Boolean combinations of basic propositions

Prior Probability

- Prior or unconditional probabilities of propositions – correspond to belief prior to arrival of any (new) evidence
  - $P(cavity) = 0.1$ and $P(sunny) = 0.75$
- Probability distribution gives values for all possible assignments
  - $P(Weather) = (0.75, 0.1, 0.1, 0.05)$ (normalized i.e., sums to 1)
- Joint probability distribution for a set of r.v.s (i.e., every sample point)
  - $P(Weather, Cavity)$ = 4×2 matrix of values
- Every question about a domain can be answered by the joint distribution because every event is a sum of sample points

Conditional Probability

- Conditional or posterior probability: probability after witnessing some evidence
  - Example: $P(cavity|toothache) = 0.8$
  - Read it as Given that $toothache$ is all I know, and not If toothache then 80% chance of cavity (does not represent causality)
- $P(Cavity|Toothache)$ = 2 element vector of 2-element vectors
- If we know more, e.g., $cavity$ is also given, then we have
  - $P(cavity|toothache, cavity) = 1$
    - The less specific belief is still valid after more evidence arrives, but is not always useful
- The new evidence may be irrelevant, allowing simplification
  - $P(cavity|toothache, KingsXIPunjabWin) = P(cavity|toothache) = 0.8$

Conditional Probability

- What is the probability of a cavity given a toothache?
- What is the probability of a cavity given the probe catches?
- Three Random Variables – $Cavity, Toothache$, and $Catch$
  - Each is either T or F

Inference by Enumeration (1)

- Start with the joint distribution

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

- For any proposition ϕ, sum the atomic events where it is true:
  - $P(\phi) = \sum_{\omega|\omega\vDash\phi} P(\omega)$

Inference by Enumeration (2)

- Start with the joint distribution

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

- For any proposition ϕ, sum the atomic events where it is true:
  - $P(\phi) = \sum_{\omega | \omega \models \phi} P(\omega)$
- $P(toothache) = 0.2$

Inference by Enumeration (3)

- Start with the joint distribution

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

- For any proposition ϕ, sum the atomic events where it is true:
  - $P(\phi) = \sum_{\omega|\omega\vDash\phi} P(\omega)$
- $P(toothache) = 0.2$
- $P(cavity \lor toothache) = 0.28$

Inference by Enumeration (4)

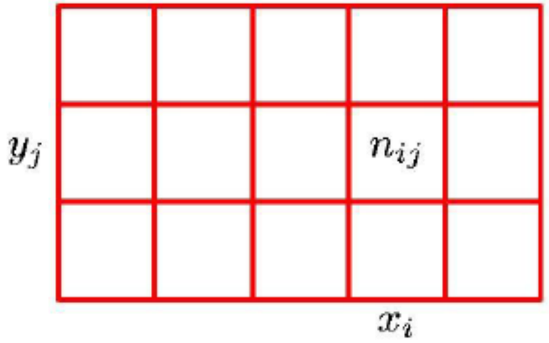
- Start with the joint distribution

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

- Can also compute conditional probabilities
- $P(\neg cavity|toothache) = 0.4$

Joint Probability

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

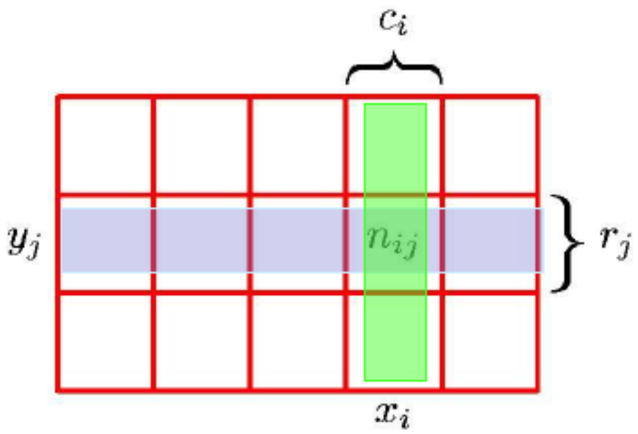


Total number of events = *N*

Marginal Probability

$$P(X = x_i) = \sum_j P(x_i, y_j) = \boxed{\frac{c_i}{N}}$$

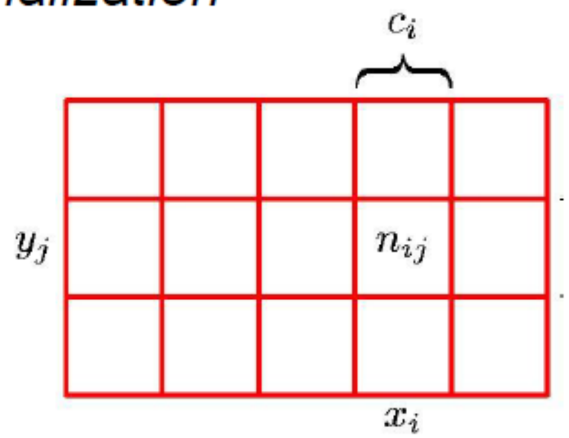
$$P(Y = y_j) = \sum_i P(x_i, y_i) = \boxed{\frac{r_j}{N}}$$



Summing out a variable is called *marginalization*

Conditional Probability

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$



Marginalization, Conditioning and Normalization

- Given full joint probabilities, marginalization or summing out is

$$P(Y) = \sum_{z \in Z} P(Y, z)$$

- Suppose we are given the conditional probabilities, then conditioning is

$$P(Y) = \sum_{z} P(Y|z)P(z)$$

- Normalization - α - normalization constant

$$P(X|y) = \alpha P(X, y)$$

$$P(x|y) = \frac{P(x, y)}{P(y)}$$

$$P(\neg x|y) = \frac{P(\neg x, y)}{P(y)}$$

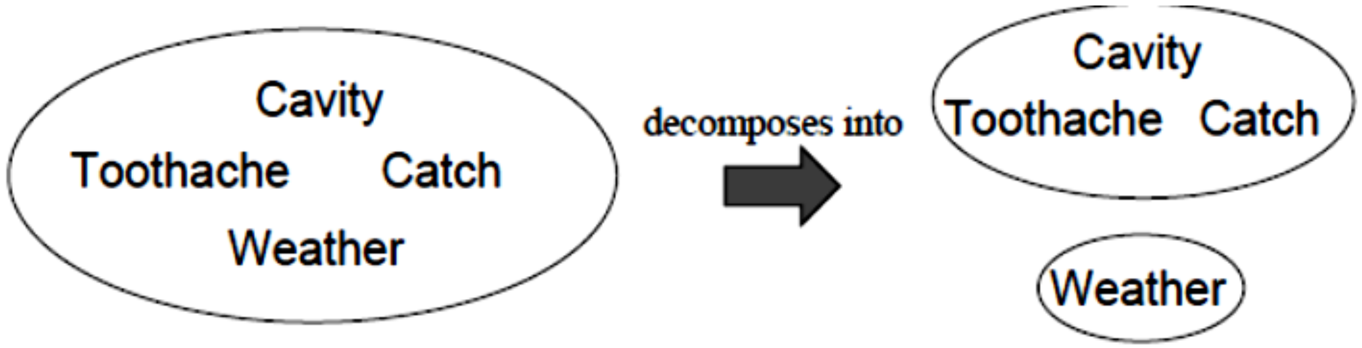
Problems with Enumeration

- Worst case time: $O(d^n)$
  - d maximum arity of the random variables
  - n - number of random variables
- Space complexity is also $O(d^n)$
  - Size of the joint distribution
- Problem: Hard/impossible to estimate all $O(d^n)$ entries of joint distribution for large problems

Independence

- A and B are independent iff

$P(A|B) = P(A)$ or $P(B|A) = P(B)$ or $P(A,B) = P(A)P(B)$



- $P(Toothache, Catch, Cavity, Weather) = P(Toothache, Catch, Cavity)P(Weather)$
- Size of the joint probability distribution table?
- Absolute independences are rare.

Conditional Independence

- $P(Toothache, Cavity, Catch)$ has $2^3 - 1 = 7$ independent entries.
- Suppose it is given that I have cavity - $Cavity = True$, then
$$P(catch|toothache, cavity) = P(catch|cavity)$$
- And similarly
$$P(catch|toothache, \neg cavity) = P(catch|\neg cavity)$$
- Which results in $Catch$ being conditionally independent of $Toothache$ given $Cavity$.
$$P(Catch|Toothache, Cavity) = P(Catch|Cavity)$$
- Equivalent statements are
$$P(Toothache|Catch, Cavity) = P(Toothache|Cavity)$$
$$P(Toothache, Catch|Cavity) = P(Toothache|Cavity)P(Catch|Cavity)$$

Conditional Independence

- Thus, factoring the joint probability distribution

$$P(Toothache, Catch, Cavity) =$$

Conditional Independence

- Thus, factoring the joint probability distribution

$$P(Toothache, Catch, Cavity)$$

$$= P(Toothache|Catch, Cavity)P(Catch, Cavity)$$

$$= P(Toothache|Cavity)P(Catch, Cavity)$$

$$= P(Toothache|Cavity)P(Catch|Cavity)P(Cavity)$$

- How many independent entries need to be stored?

Conditional Independence

- Thus, factoring the joint probability distribution

$P(Toothache, Catch, Cavity)$

$= P(Toothache|Catch, Cavity)P(Catch, Cavity)$

$= P(Toothache|Cavity)P(Catch, Cavity)$

$= P(Toothache|Cavity)P(Catch|Cavity)P(Cavity)$

- How many independent entries need to be stored? 5
- Conditional Independence reduces the size of the representation in JPT that is exponential in N to linear in N.

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Bayesian Networks

Material adapted from Dan Klein

Probabilistic Models

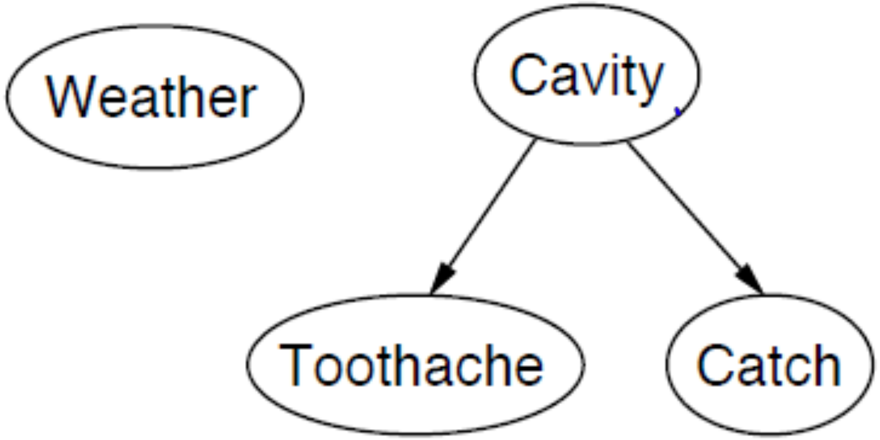
- Models that describe how a portion of the real-world works
  - Simplifications of the real world
- What do we do with probabilistic models
  - Agents reason about unknown variables, given evidence
    - Reason about unknown variables
      - Explanation (diagnostic reasoning)
      - Prediction (causal reasoning)
    - Value of information

Bayesian Networks

- Graphical notation for conditional independence assumptions
  - Compact specification of full joint distributions
- Syntax
  - One node per random variable
  - A directed edge from one node to another (influences)
  - A conditional distribution at each node given its parents - $P\left(X_i | Parents(X_i)\right)$
    - In the simplest case, represented as a conditional probability distribution table (CPT)
  - Directed Acyclic Graph

Example (1)

- Topology of the network encodes the conditional independence assumptions

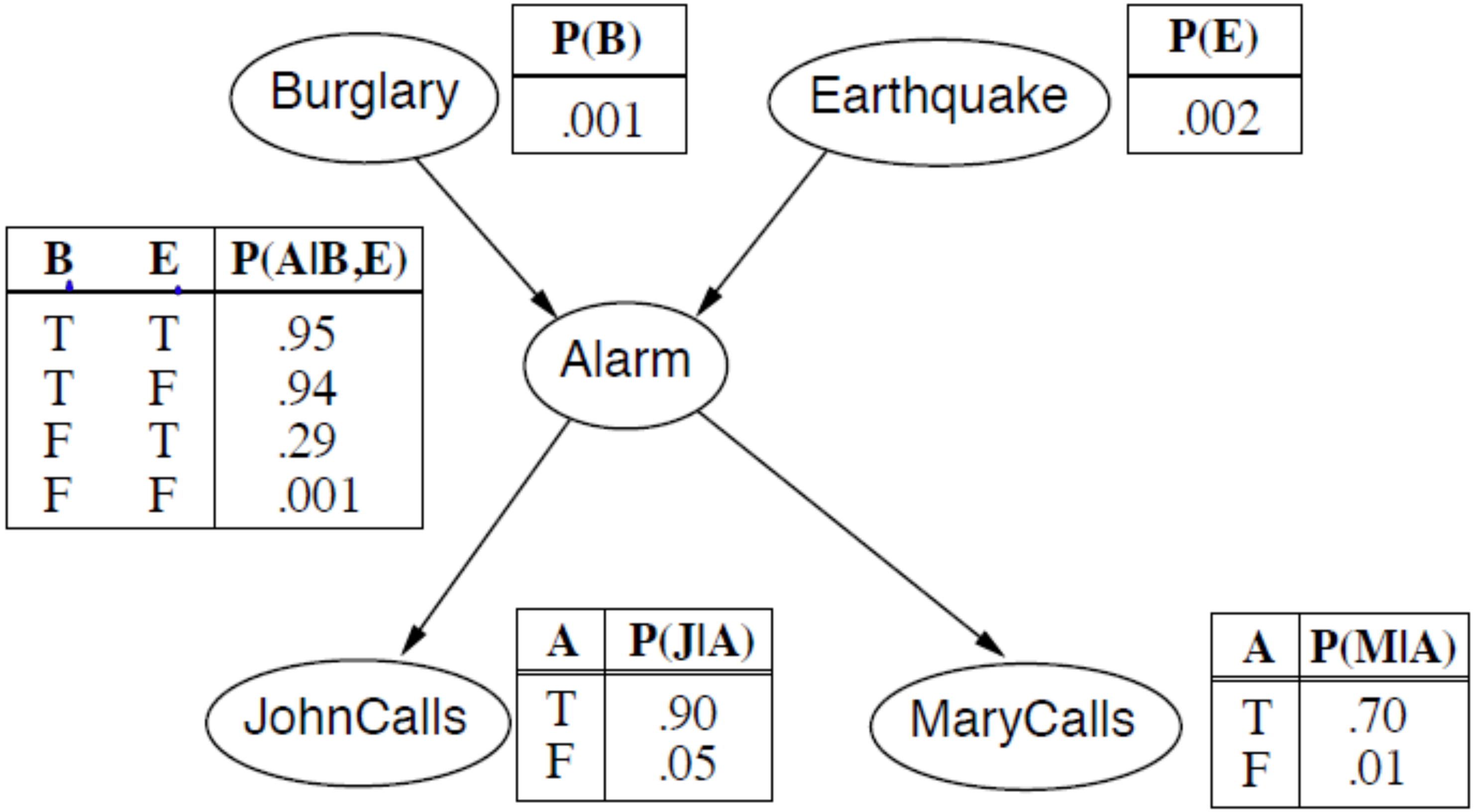


- Weather is independent of other variables
- $Toothache$ and $Catch$ are conditionally independent given $Cavity$

Example (2)

- I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes the alarm is set off by earthquakes. Is there a burglar?
- Variables – $Burglar, Earthquake, Alarm, JohnCalls, MaryCalls$
- Network topology reflects the "casual" knowledge
  - Burglar can set off the alarm
  - Earthquake can set off the alarm
  - The alarm can cause John to call
  - The alarm can cause Mary to call

Example (2)



P(B)
.001

P(E)
.002

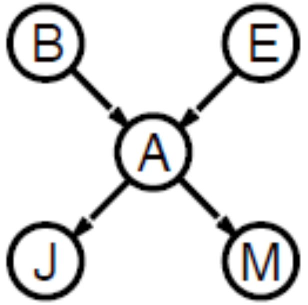
B	E	P(A|B,E)
T	T	.95
T	F	.94
F	T	.29
F	F	.001

A	P(J|A)
T	.90
F	.05

A	P(M|A)
T	.70
F	.01

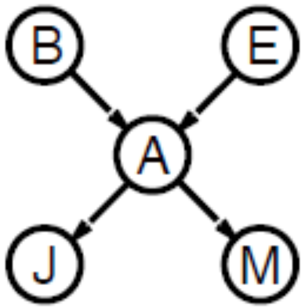
Compactness

- A CPT for X_i with k Boolean parents has 2^k rows for the combination of parent values.
- Each row requires one number p for $X_i = True$
  - $X_i = False$ can be computed from p
- If each variable has no more than k parents, the complete network requires $O(n2^k)$ entries
  - Thus, the representation grows linearly in n, in contrast to full joint probability distribution table.
- For the burglary net



Global Semantics

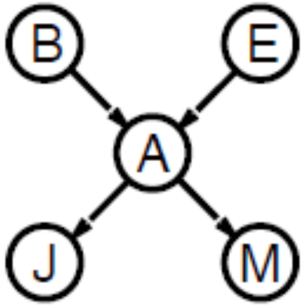
- Define the full joint distribution as a product of the local conditional distributions
- $P(X_1, \dots X_N) = \prod_{i=1}^{N} P\big(X_i | Parents(X_i)\big)$
- For example
- $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) =$



Global Semantics

- Define the full joint distribution as a product of the local conditional distributions
- $P(X_1, \dots X_N) = \prod_{i=1}^{N} P\big(X_i | Parents(X_i)\big)$
- For example
- $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$

$$= P(j|a)P(m|a)P(a|\neg b, \neg e)P(\neg b)P(\neg e)$$
$$= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 \approx 0.0006$$



Bayes Nets – Causality?

- When BNs reflect the true causal patterns:
  - Often simpler (nodes have fewer parents)
  - Often easier to think about
  - Often easier to elicit from experts
- BNs need not actually be causal
  - Sometimes no causal net exists over the domain
  - End up with arrows that reflect correlation, not causation
- What do the arrows really mean
  - Topology may happen to encode the causal structure
  - *Topology really encodes conditional independence*.

Constructing Bayesian Networks

- A series of locally testable assertions of conditional independence guarantees global semantics (full joint)
- Choose an ordering of the variables - $X_1, \ldots, X_N$
- For $i = 1, \ldots, N$
  - Add variable X_i to the network
  - Select parents from $X_1, \ldots, X_{i-1}$ such that

$$P\left(X_i | Parents(X_i)\right) = P(X_i | X_1, \ldots, X_{i-1})$$

- This choice guarantees global semantics

$$P(X_1, \ldots, X_N) = \prod_{i=1}^{N} P(X_i | X_1, \ldots, X_{i-1}) = \prod_{i=1}^{N} P(X_i | Parents(X_i))$$

Example

- Suppose we choose the ordering M, J, A, B, E

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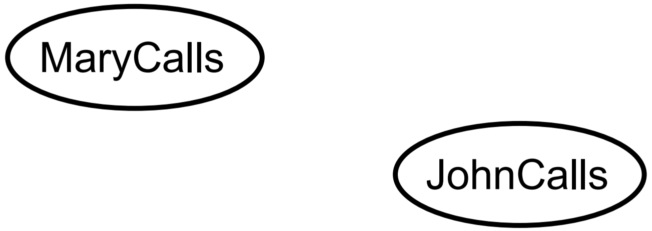

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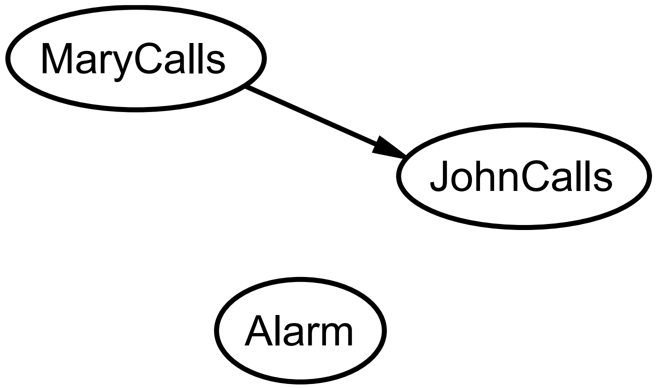

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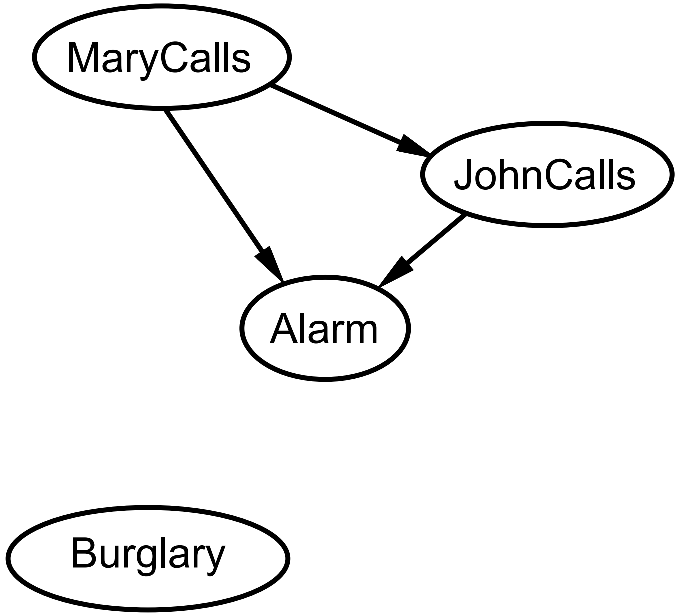

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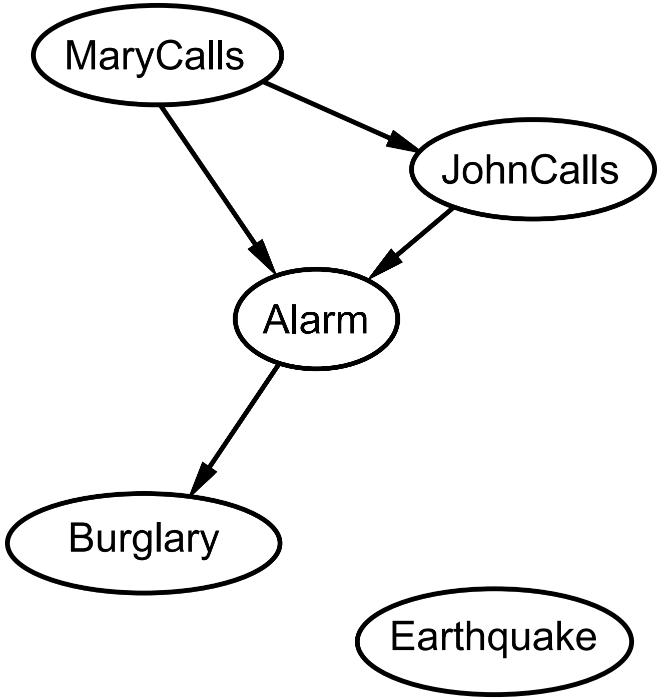

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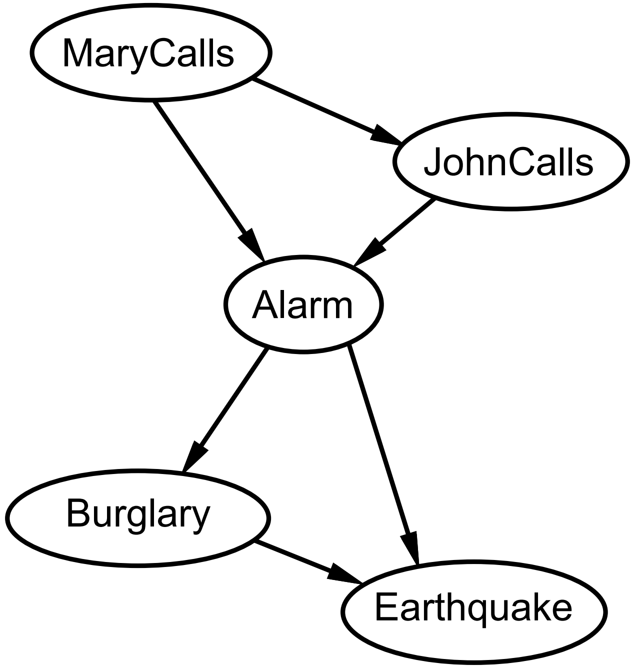
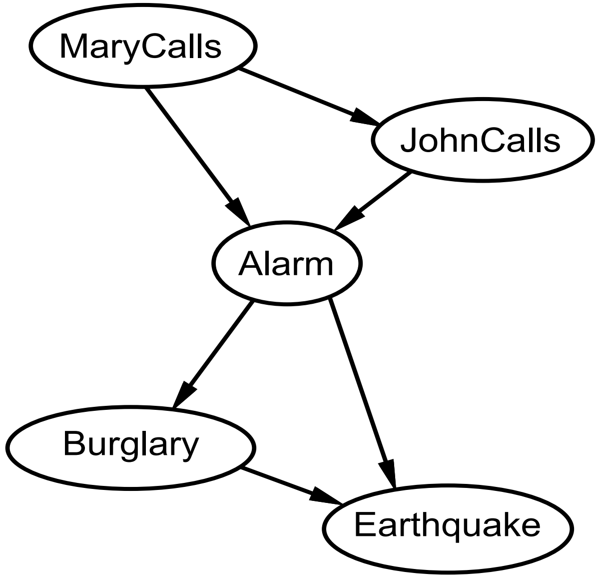


Example

- Deciding conditional independence is hard in non-causal directions
  - Causal models and conditional independencies seems hardwired in humans
- Assessing conditional independencies in non-causal directions is hard
- Size of the network?



Bayes Nets: Conditional Independence

- Fundamental Assumption

$$P(x_i|x_1,\dots,x_{i-1}) = P(x_i|Parents(X_i))$$

- Beyond the Bayes net Chain rule conditional independence assumptions,
  - Additional conditional independencies, that can be read off the graph.
  - Example

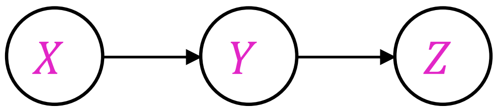
$$W \rightarrow X \rightarrow Y \rightarrow Z$$

CI assumptions directly from simplifications in chain rule:

Other implied CI assumptions

Bayes Nets: Conditional Independence (2)

- Given a BN, a common yet important question would be
  - Are two nodes independent given certain evidence?
    - If yes, prove using probabilistic algebra
    - If no, give a counter example.
  - Consider the following example
    - Are X and Z independent?

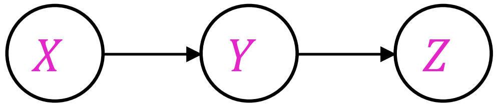


D-Separation

- Answer independence queries regarding variables in a BN.
  - Study independence properties for triples
    - Causal chain, common cause and common effect
  - Analyze complex cases in terms of member triples
  - D-Separation: a condition/algorithm for answering these queries.

D-Separation – Causal Chains

- X – Transfer money, Y- non-zero balance, Z – withdraw money



$$P(X,Y,Z) = P(X)P(Y|X)P(Z|Y)$$

D-Separation – Causal Chains

- X – Transfer money, Y - non-zero balance, Z – withdraw money

$$X \rightarrow Y \rightarrow Z$$

$$P(X,Y,Z) = P(X)P(Y|X)P(Z|Y)$$

- Is it guaranteed that X is independent of Z?
  - No! - Proof by counter example
    - Consider the case- Transferring money causes non-zero balance causes withdrawal

$P(y|x) = 1, P(\neg y|\neg x) = 1$

$P(z|y) = 1, P(\neg z|\neg y) = 1$

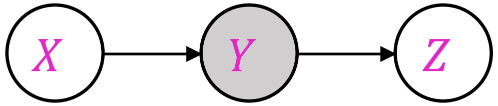
And the evidence

$P(x) = 0.7$

$$P(x,z) \neq P(x)P(z)$$

D-Separation – Causal Chains

- X – Transfer money, Y- non-zero balance, Z – withdraw money



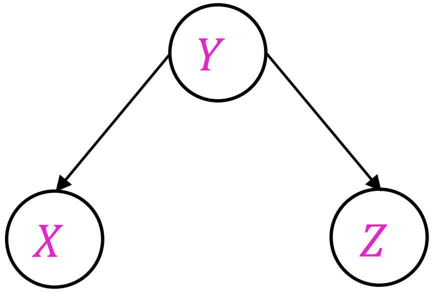
$$P(X,Y,Z) = P(X)P(Y|X)P(Z|Y)$$

- Is it guaranteed that X is independent of Z given Y?

- Evidence along the chain “blocks” the influence

D-Separation – Common Cause

- X – thin attendance, Y- project is due, Z – students are busy



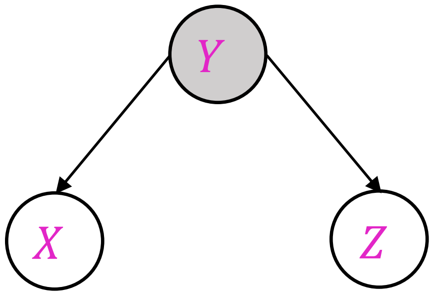
$$P(X,Y,Z) = P(Y)P(X|Y)P(Z|Y)$$

- Is it guaranteed that X is independent of Z?
  - No! - Proof by counter example
    - Consider the scenario Project due causes both thin attendance and students busy

$P(x|y) = 1, P(\neg x|\neg y) = 1$
$P(z|y) = 1, P(\neg z|\neg y) = 1$
And the evidence
$P(y) = 0.7$
$P(x,z) \neq P(x)P(z)$

D-Separation – Common Cause

- X – thin attendance, Y- project is due, Z – students are busy



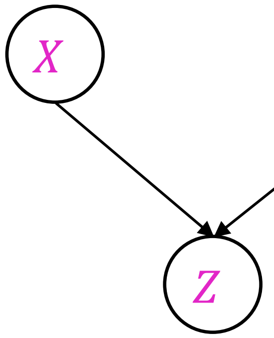
$$P(X,Y,Z) = P(Y)P(X|Y)P(Z|Y)$$

- Is it guaranteed that X is independent of Z given Y?

- Observing the cause blocks the influence between effects

D-Separation – Common Effect

- X – RoadRepair, Y- Cricket Match, Z - Traffic

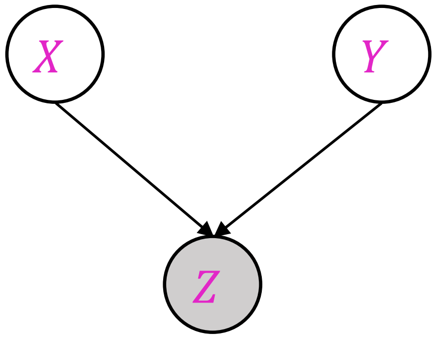


$$P(X,Y,Z) = P(X)P(Y)P(Z|X,Y)$$

- Is it guaranteed that X is independent of Y?

D-Separation – Common Effect

- X – RoadRepair, Y- Cricket Match, Z - Traffic

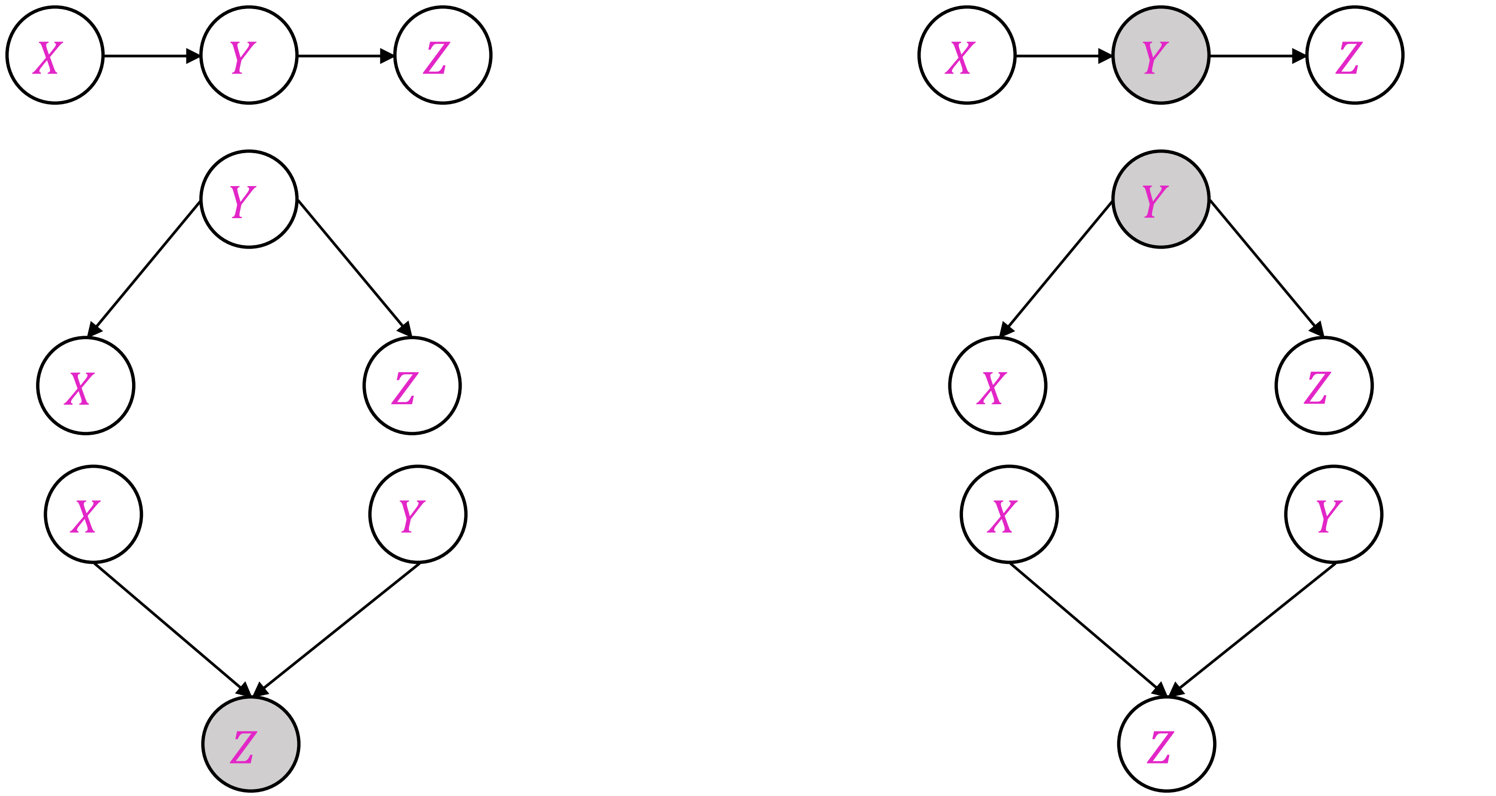


$$P(X,Y,Z) = P(X)P(Y)P(Z|X,Y)$$

- Is it guaranteed that X is independent of Y given Z?
  - No, seeing traffic puts the roadrepair and the cricket match as explanation
- This is different from the other cases
  - Observing an effect activates influence between possible causes

D-Separation – Active/Inactive Paths

- A path is active if each triple in the path is active - dependencies between nodes.

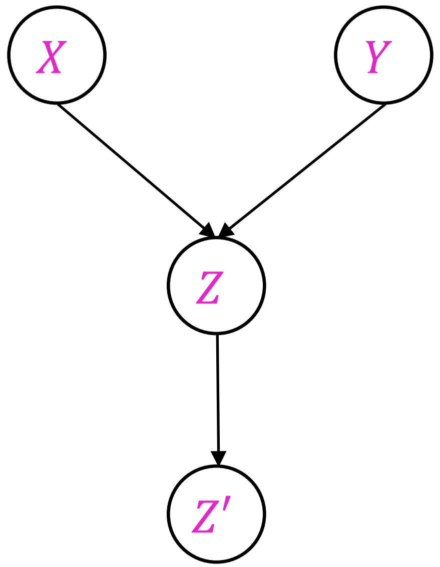


D-separation

- Given a BN, are two variables X and Y independent (given evidence Z)?
  - Yes if X and Y are ‘d-separated’ by Z
  - All paths are inactive implies independence
    - A single inactive segment is sufficient to block the path
  - If one or more paths are active, then independence is not guaranteed
- Analyze the graph
  - Check for the three canonical forms all through the path

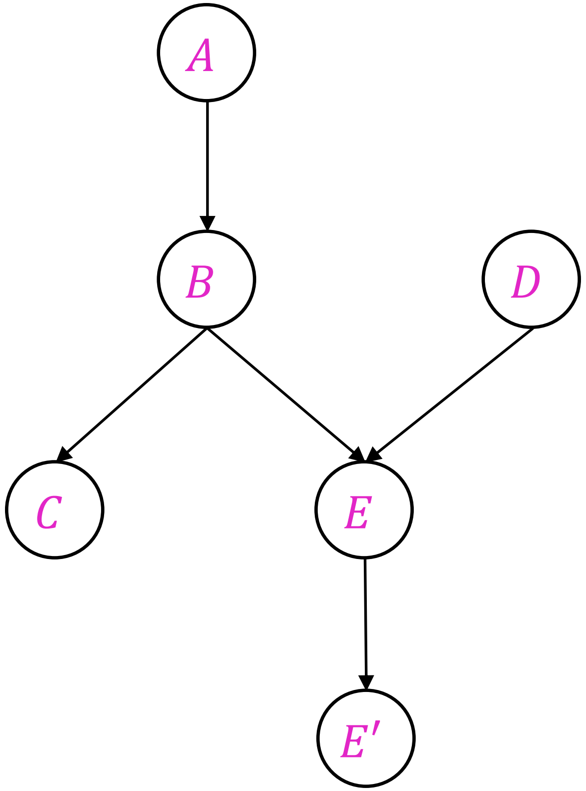
Example

- X independent of Y
- X independent of Y given Z
- X independent of Y given Z'

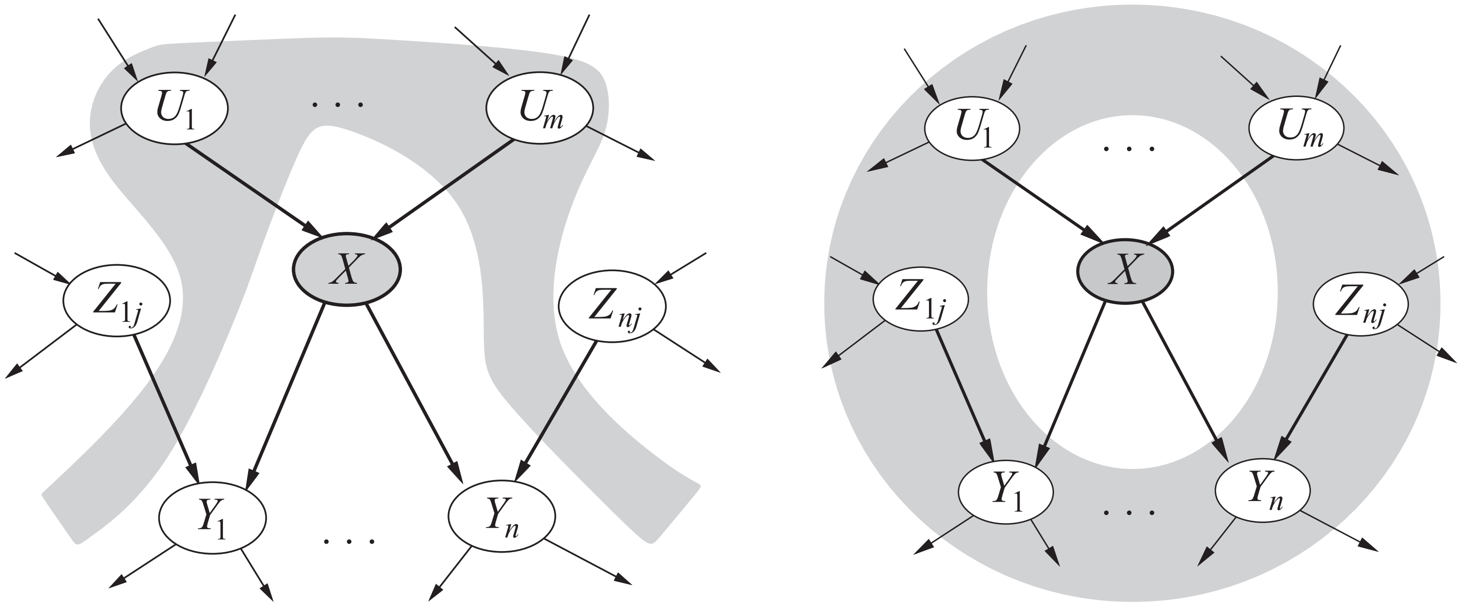


Example (2)

- ❑ C independent of D
- ❑ A independent of E’ given E
- ❑ A independent of D
- ❑ A independent of D given E
- ❑ A independent of D given B, E



Conditional Independence in BNs – Markov Blanket



Outline

- Probability theory - basics
- Probabilistic Inference
  - Inference by enumeration
- Conditional Independence
- Bayesian Networks
- Inference in Bayesian Networks
  - Exact Inference
  - Approximate Inference

Probabilistic Inference in BNs

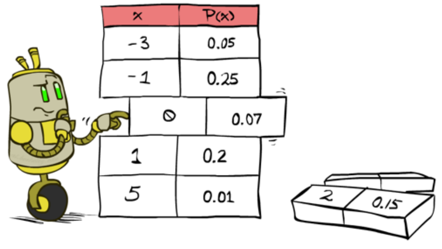
- Graphical independence representation yields efficient inference schemes
- We generally want to compute
  - $P(X|E)$, where E is the evidence from the sensory measurements (known values for variables)
  - Sometimes, we may want to compute just $P(X)$
- Three approaches
  - Enumeration
  - Variable Elimination
  - Approximate Inference
    - sampling

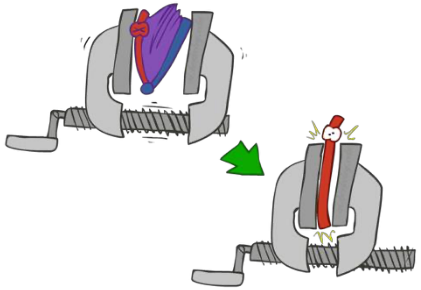
Inference by Enumeration

- General case - the set of random variables $X_1, \dots, X_{\mathrm{n}}$ is broken into three categories
  - Evidence variables - $E_1, \dots, E_k = e_1, \dots, e_k$
  - Query variables - $Q_1, \dots, Q_L = Q$
  - Hidden variables - $H_1, \dots, H_{\mathrm{r}}$
- We want to estimate - $P(Q | e_1, \dots, e_k)$

Inference by Enumeration

- We want to estimate - $P(Q|e_1, \dots, e_k)$
- Step 1 – select the consistent entries with the evidence
- Step 2 - sum of the hidden variables to get the joint of Query and evidence
  - $P(Q, e_1, \dots, e_k) = \sum_{H_1, \dots H_r} P(Q, h_1, \dots, h_r, e_1, \dots, e_k)$
- Step 3 – Normalize
  - $Z = \sum_q P(Q, e_1, \dots, e_k)$
  - $P(Q|e_1, \dots, e_k) = \frac{1}{Z} P(Q, e_1, \dots, e_k)$

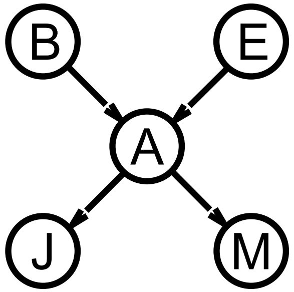




$$\times \frac{1}{Z}$$

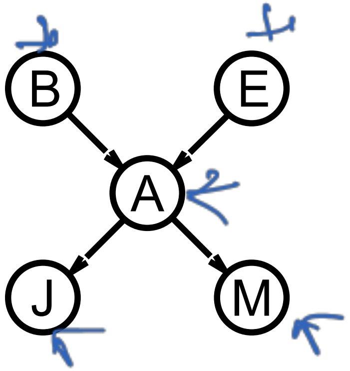
Inference by Enumeration

- Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.
- Consider the simple query on the burglary network
  - $P(B|j,m)$



Inference by Enumeration (2)

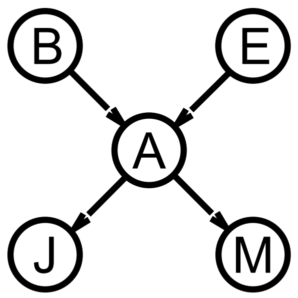
- Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.
- Consider the simple query on the burglary network
  - $P(B|j,m) = P(B,j,m)/P(j,m)$
  - $= \frac{1}{Z} P(B,j,m)$
  - $= \frac{1}{Z} \sum_e \sum_a P(B,e,a,j,m)$



- Rewrite full joint entries using product of the CPT entries

Inference by Enumeration (3)

- Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.
- Consider the simple query on the burglary network
  - $P(B|j, m) = P(B, j, m)/P(j, m)$
  - $= \frac{1}{Z} P(B, j, m)$
  - $= \frac{1}{Z} \sum_e \sum_a P(B, e, a, j, m)$



- Rewrite full joint entries using product of the CPT entries
  - $= \frac{1}{Z} P(B) \sum_e P(e) \sum_a P(a|B, e)\, P(j|a) P(m|a)$

Evaluation Tree

- Recursive depth first enumeration
  - $= \frac{1}{Z} P(B) \sum_e P(e) \sum_a P(a|B,e)\, P(j|a) P(m|a)$